Construction of the Central Moment Shift Operator $\boldsymbol{T}(\tilde{\boldsymbol{u}})$ for D3Q19 MRT Lattice Boltzmann Methods: Lattice-Reduced Polynomial Approach

Technical Report — GILBM Project

March 2026

Abstract

We present a complete, self-contained derivation of the shift operator $\boldsymbol{T}(\tilde{\boldsymbol{u}})$ that transforms raw moments to central moments in the D3Q19 Multiple-Relaxation-Time (MRT) Lattice Boltzmann framework. The shift operator is defined via Dubois et al. (2015) as $\boldsymbol{T}(\tilde{\boldsymbol{u}}) = \boldsymbol{M}(\tilde{\boldsymbol{u}})\,\boldsymbol{M}^{-1}(0)$, where $\boldsymbol{M}(\tilde{\boldsymbol{u}})$ evaluates the d'Humières polynomial basis at shifted lattice velocities $\boldsymbol{e}_i - \tilde{\boldsymbol{u}}$. We show that the d'Humières basis polynomials for D3Q19 contain degree-4 terms whose continuous polynomial extension is *not unique* on the lattice. We identify two choices — the "Dubois continuous" form and the "lattice-reduced" form — and prove that only the latter yields an algebraically exact group structure $\boldsymbol{T}(\boldsymbol{u})\boldsymbol{T}(-\boldsymbol{u}) = \boldsymbol{I}$. Both forms produce identical stress-mode central moments and hence the same Navier–Stokes viscosity (Corollary 2.1 of Dubois et al.). Full closed-form expressions for the 19×19 $\boldsymbol{T}(\tilde{\boldsymbol{u}})$ and $\boldsymbol{T}^{-1}(\tilde{\boldsymbol{u}})$ are given.

Contents

1 Introduction and Motivation **3**

2 The D3Q19 Lattice and d'Humières Basis **3**
2.1 Lattice velocities . . . 3
2.2 The lattice reduction identity . . . 3
2.3 The d'Humières polynomial basis . . . 3

3 Complete D3Q19 Polynomial Basis: Two Versions **4**
3.1 Polynomials with unique extension (13 rows) . . . 4
3.2 Polynomials with ambiguous extension (6 rows) . . . 5

4 Construction of $\boldsymbol{T}(\tilde{\boldsymbol{u}})$ **5**
4.1 General definition (Dubois et al. 2015) . . . 5
4.2 Why polynomial choice matters . . . 5
4.3 Block-triangular structure . . . 5

5 The Lattice-Reduced Shift Operator: Closed Form **6**
5.1 Conserved moments (degree 0 and 1, rows 0,3,5,7) . . . 6
5.2 Stress moments (degree 2, rows 9,11,13,14,15) . . . 6
5.3 Energy moment (degree 2, row 1) . . . 6
5.4 Third-order moments (rows 16,17,18) . . . 7
5.5 Energy-flux moments (degree 3/4, rows 4,6,8) — **version-dependent** . . . 7
5.6 Ghost/energy-squared moments (degree 4, rows 2,10,12) — **version-dependent** 7

6 Inverse Shift Operator $\boldsymbol{T}^{-1}(\tilde{\boldsymbol{u}})$ **7**
6.1 Lattice-reduced version: exact inverse via sign flip 7
6.2 Dubois continuous version: approximate inverse 8

7 Comparison of the Two Versions **8**
7.1 What is identical 8
7.2 What differs 8
7.3 Correction structure 9
7.4 Summary table 9

8 Physical Interpretation **9**

9 Implementation Notes **9**
9.1 Efficient CUDA implementation 9
9.2 Compile-time switch 10

10 Conclusion **10**

A Full Expressions for All Remaining Rows **10**
A.1 Row 10: $3\pi_{xx}$ (energy $\times$ stress) 10
A.2 Row 12: π_{ww} (energy $\times$ stress) 11
A.3 Row 6: q_y (energy flux, y-direction) 11
A.4 Row 8: q_z (energy flux, z-direction) 11

B Dubois Correction Terms $\Delta\boldsymbol{T}$ **11**

1 Introduction and Motivation

The MRT-CM (Central Moment) collision operator for lattice Boltzmann methods relaxes the non-equilibrium distribution in a frame co-moving with the local fluid velocity, thereby improving Galilean invariance compared to standard MRT relaxation in the rest frame. The collision step takes the form

$$f_i^* = f_i - \left(\boldsymbol{M}^{-1}\,\boldsymbol{T}^{-1}(\boldsymbol{u})\,\boldsymbol{S}\,\boldsymbol{T}(\boldsymbol{u})\,\boldsymbol{M}\right)_{ij}(f_j - f_j^{\mathrm{eq}}), \tag{1}$$

where $\boldsymbol{M}$ is the moment transformation matrix, $\boldsymbol{S} = \mathrm{diag}(s_0, \ldots, s_{18})$ is the diagonal relaxation matrix, and $\boldsymbol{T}(\boldsymbol{u})$ is the *shift operator* that converts raw moments to central moments.

Dubois et al. (2015) introduced a general framework: given a polynomial basis $\{P_0, \ldots, P_{q-1}\}$, the shifted moment matrix is $M_{kj}(\tilde{\boldsymbol{u}}) = P_k(\boldsymbol{v}_j - \tilde{\boldsymbol{u}})$, and the shift operator is

$$\boldsymbol{T}(\tilde{\boldsymbol{u}}) = \boldsymbol{M}(\tilde{\boldsymbol{u}})\,\boldsymbol{M}^{-1}(0). \tag{2}$$

However, Dubois et al. only derived *explicit* closed-form expressions for D2Q9. **No explicit construction for D3Q19 was given in their paper.** This report fills that gap: we derive the complete 19×19 shift operator for D3Q19 using the d'Humières (2002) polynomial basis, and we identify a critical subtlety — the polynomial extension ambiguity — that must be resolved to obtain correct algebraic properties.

2 The D3Q19 Lattice and d'Humières Basis

2.1 Lattice velocities

The D3Q19 lattice has velocity vectors $\boldsymbol{e}_i = (e_{ix}, e_{iy}, e_{iz})$ with components in $\{-1, 0, 1\}$:

$$\begin{aligned}
&\boldsymbol{e}_0 = (0,0,0),\\
&\boldsymbol{e}_{1,2} = (\pm1,0,0), \quad \boldsymbol{e}_{3,4} = (0,\pm1,0), \quad \boldsymbol{e}_{5,6} = (0,0,\pm1),\\
&\boldsymbol{e}_{7\text{–}10} = (\pm1,\pm1,0), \quad \boldsymbol{e}_{11\text{–}14} = (\pm1,0,\pm1), \quad \boldsymbol{e}_{15\text{–}18} = (0,\pm1,\pm1).
\end{aligned} \tag{3}$$

2.2 The lattice reduction identity

Since all velocity components satisfy $e_{i\alpha} \in \{-1, 0, 1\}$, we have the fundamental identities:

$$\boxed{e_{i\alpha}^{2n} = e_{i\alpha}^2, \qquad e_{i\alpha}^{2n+1} = e_{i\alpha}, \qquad \forall\, n \geq 1.} \tag{4}$$

This means that on the lattice, $x^4 = x^2$, $x^3 = x$, $x^6 = x^2$, etc. Consequently, any polynomial evaluated on the lattice can be reduced to a polynomial involving only the **19 independent monomials**:

$$\{1,\ x,\ y,\ z,\ x^2,\ y^2,\ z^2,\ xy,\ xz,\ yz,\ x^2y,\ x^2z,\ xy^2,\ y^2z,\ xz^2,\ yz^2,\ x^2y^2,\ x^2z^2,\ y^2z^2\}. \tag{5}$$

Remark 2.1. *These 19 monomials span $\mathbb{R}[\boldsymbol{e}]|_{D3Q19}$, the space of all polynomial functions restricted to the 19 lattice points. Any polynomial in x, y, z evaluated on D3Q19 is equivalent to a unique linear combination of these 19 monomials.*

2.3 The d'Humières polynomial basis

d'Humières et al. (2002) defined the moment matrix $\boldsymbol{M}$ by its action on lattice velocities: $M_{k,i} = P_k(e_{ix}, e_{iy}, e_{iz})$, where $\{P_k\}_{k=0}^{18}$ are polynomials chosen for physical interpretability. On the lattice, these polynomials are *uniquely determined* by the 19 numerical values in each row of $\boldsymbol{M}$. However, their *continuous polynomial extensions* (i.e., their expressions as functions of continuous variables $X, Y, Z \in \mathbb{R}$) are **not unique** due to the lattice reduction identities (4).

We illustrate with the energy-squared moment P_2 (row index 2). The numerical values of $\boldsymbol{M}$ give:

$$\boldsymbol{M}[2,:] = (12,\ -4,\ -4,\ -4,\ -4,\ -4,\ -4,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1,\ 1).$$

Two valid continuous polynomial extensions that reproduce these values on the lattice are:

(A) Dubois continuous form (keeping $r^4 = (X^2+Y^2+Z^2)^2$ as a true degree-4 polynomial):

$$P_2^{\text{cont}}(X,Y,Z) = \tfrac{21}{2}\, r^4 - \tfrac{53}{2}\, r^2 + 12. \tag{6}$$

(B) Lattice-reduced form (applying $X^4 \to X^2$, etc.):

$$P_2^{\text{red}}(X,Y,Z) = -16\, r^2 + 21\,(X^2Y^2 + X^2Z^2 + Y^2Z^2) + 12. \tag{7}$$

Proposition 2.1. *$P_2^{cont}(\boldsymbol{e}_i) = P_2^{red}(\boldsymbol{e}_i)$ for all $i = 0, \ldots, 18$.*

Proof. On the lattice, $X^4 = X^2$, so $r^4 = (X^2+Y^2+Z^2)^2 = X^2+Y^2+Z^2+2(X^2Y^2+X^2Z^2+Y^2Z^2)$ (using $X^4 = X^2$). Substituting into (6): $\frac{21}{2}[r^2+2(X^2Y^2+X^2Z^2+Y^2Z^2)] - \frac{53}{2}\, r^2 + 12 = -16\, r^2 + 21(X^2Y^2+X^2Z^2+Y^2Z^2) + 12$. This is exactly (7). □

3 Complete D3Q19 Polynomial Basis: Two Versions

We now list all 19 polynomials in both forms. The rows are grouped by physical type. Polynomials of degree ≤ 2 have unique extensions (no ambiguity); the difference arises only in the 6 polynomials containing degree-3 or degree-4 terms.

3.1 Polynomials with unique extension (13 rows)

These polynomials have degree ≤ 3 using only the 19 independent monomials, so both versions agree:

k	Label	Type	$P_k(X,Y,Z)$
0	ρ (density)	conserved	1
1	e (energy)	ghost	$19\, r^2 - 30$
3	j_x	conserved	X
5	j_y	conserved	Y
7	j_z	conserved	Z
9	$3p_{xx}$ (stress)	stress	$2X^2 - Y^2 - Z^2$
11	p_{ww} (stress)	stress	$Y^2 - Z^2$
13	p_{xy} (stress)	stress	XY
14	p_{yz} (stress)	stress	YZ
15	p_{xz} (stress)	stress	XZ
16	m_x (3rd)	3rd-order	$X(Y^2 - Z^2)$
17	m_y (3rd)	3rd-order	$Y(Z^2 - X^2)$
18	m_z (3rd)	3rd-order	$Z(X^2 - Y^2)$

Here $r^2 \equiv X^2 + Y^2 + Z^2$. These are degree ≤ 3 and involve only the 19 independent monomials (5).

3.2 Polynomials with ambiguous extension (6 rows)

k	**Dubois Continuous** P_k^{cont}	**Lattice-Reduced** P_k^{red}
2	$\frac{21}{2}\,r^4 - \frac{53}{2}\,r^2 + 12$	$-16\,r^2 + 21(X^2Y^2{+}X^2Z^2{+}Y^2Z^2) + 12$
4	$5X\,r^2 - 9X$ (keeps X^3)	$-4X + 5XY^2 + 5XZ^2$ ($X^3 \to X$)
6	$5Y\,r^2 - 9Y$	$-4Y + 5YX^2 + 5YZ^2$
8	$5Z\,r^2 - 9Z$	$-4Z + 5ZX^2 + 5ZY^2$
10	$(2X^2{-}Y^2{-}Z^2)(3r^2{-}5)$ (keeps X^4, X^2Y^2 as deg-4)	$-4X^2{+}2Y^2{+}2Z^2 + 3(X^2Y^2{+}X^2Z^2) - 6Y^2Z^2$
12	$(Y^2{-}Z^2)(3r^2{-}5)$	$-2Y^2{+}2Z^2 + 3X^2Y^2 - 3X^2Z^2$

Remark 3.1. *On the D3Q19 lattice points, every $P_k^{cont}(\boldsymbol{e}_i) = P_k^{red}(\boldsymbol{e}_i)$ exactly, by the same argument as Proposition 2.1. The two forms are indistinguishable when evaluated at $\boldsymbol{e}_i$. They differ only when evaluated at shifted points $\boldsymbol{e}_i - \tilde{\boldsymbol{u}}$, which is precisely what happens in the construction of $\boldsymbol{T}(\tilde{\boldsymbol{u}})$.*

4 Construction of $\boldsymbol{T}(\tilde{\boldsymbol{u}})$

4.1 General definition (Dubois et al. 2015)

Given a polynomial basis $\{P_k\}$, define the shifted moment matrix:

$$M_{kj}(\tilde{\boldsymbol{u}}) = P_k(\boldsymbol{e}_j - \tilde{\boldsymbol{u}}), \qquad 0 \le k, j \le 18. \tag{8}$$

The shift operator is:

$$\boldsymbol{T}(\tilde{\boldsymbol{u}}) = \boldsymbol{M}(\tilde{\boldsymbol{u}})\,\boldsymbol{M}^{-1}(0). \tag{9}$$

This matrix converts "fixed" raw moments $\boldsymbol{m}(0) = \boldsymbol{M}(0)\,\boldsymbol{f}$ into "shifted" moments $\boldsymbol{m}(\tilde{\boldsymbol{u}}) = \boldsymbol{T}(\tilde{\boldsymbol{u}})\,\boldsymbol{m}(0)$.

4.2 Why polynomial choice matters

The shifted point is $\boldsymbol{e}_j - \tilde{\boldsymbol{u}}$, which is *not* a lattice point (since $\tilde{\boldsymbol{u}} \in \mathbb{R}^3$ is continuous). At this point, $X^4 \neq X^2$ in general. Therefore, $P_k^{\text{cont}}(\boldsymbol{e}_j - \tilde{\boldsymbol{u}}) \neq P_k^{\text{red}}(\boldsymbol{e}_j - \tilde{\boldsymbol{u}})$ for rows $k \in \{2, 4, 6, 8, 10, 12\}$, leading to two different shift operators:

$$\boldsymbol{T}^{\text{cont}}(\tilde{\boldsymbol{u}}) \neq \boldsymbol{T}^{\text{red}}(\tilde{\boldsymbol{u}}) \qquad \text{for } \tilde{\boldsymbol{u}} \neq \boldsymbol{0}. \tag{10}$$

4.3 Block-triangular structure

Both versions of $\boldsymbol{T}(\tilde{\boldsymbol{u}})$ share the same block-triangular structure (by degree ordering). The d'Humières polynomials have degrees:

Degree	Row indices	Count
0	$\{0\}$	1
1	$\{3, 5, 7\}$	3
2	$\{1, 9, 11, 13, 14, 15\}$	6
3	$\{4, 6, 8, 16, 17, 18\}$	6
4	$\{2, 10, 12\}$	3

Since shifting $P_k(\boldsymbol{X} - \tilde{\boldsymbol{u}})$ introduces terms of degree $\leq \deg(P_k)$, the matrix $\boldsymbol{T}$ is block lower-triangular with respect to this degree ordering, with identity blocks on the diagonal.

5 The Lattice-Reduced Shift Operator: Closed Form

We now derive the complete closed-form expressions. Denote $\tilde{\boldsymbol{u}} = (\tilde{u}_x, \tilde{u}_y, \tilde{u}_z)$ and abbreviate $u_x^2 = \tilde{u}_x^2$, $u^2 = \tilde{u}_x^2 + \tilde{u}_y^2 + \tilde{u}_z^2$, etc.

The shift operator acts on the raw moment vector $\boldsymbol{m} = (m_0, m_1, \ldots, m_{18})^\top$ to produce the central moment vector $\boldsymbol{k} = (k_0, k_1, \ldots, k_{18})^\top$:

$$\boldsymbol{k} = \boldsymbol{T}(\tilde{\boldsymbol{u}})\, \boldsymbol{m}. \tag{11}$$

5.1 Conserved moments (degree 0 and 1, rows 0,3,5,7)

These are independent of the polynomial extension choice:

$$k_0 = m_0, \tag{12}$$
$$k_3 = -\tilde{u}_x\, m_0 + m_3, \tag{13}$$
$$k_5 = -\tilde{u}_y\, m_0 + m_5, \tag{14}$$
$$k_7 = -\tilde{u}_z\, m_0 + m_7. \tag{15}$$

Remark 5.1. *For $\boldsymbol{f} = f^{\mathrm{eq}}$, we have $m_0 = \rho$ and $m_{3,5,7} = \rho\, \tilde{u}_{x,y,z}$, giving $k_0 = \rho$ and $k_{3,5,7} = 0$, confirming that central moments of the equilibrium have zero momentum.*

5.2 Stress moments (degree 2, rows 9,11,13,14,15)

Also independent of the polynomial extension (degree ≤ 2, no ambiguity):

$$k_9 = (2\tilde{u}_x^2 - \tilde{u}_y^2 - \tilde{u}_z^2)\, m_0 - 4\tilde{u}_x\, m_3 + 2\tilde{u}_y\, m_5 + 2\tilde{u}_z\, m_7 + m_9, \tag{16}$$
$$k_{11} = (\tilde{u}_y^2 - \tilde{u}_z^2)\, m_0 - 2\tilde{u}_y\, m_5 + 2\tilde{u}_z\, m_7 + m_{11}, \tag{17}$$
$$k_{13} = \tilde{u}_x\tilde{u}_y\, m_0 - \tilde{u}_y\, m_3 - \tilde{u}_x\, m_5 + m_{13}, \tag{18}$$
$$k_{14} = \tilde{u}_y\tilde{u}_z\, m_0 - \tilde{u}_z\, m_5 - \tilde{u}_y\, m_7 + m_{14}, \tag{19}$$
$$k_{15} = \tilde{u}_x\tilde{u}_z\, m_0 - \tilde{u}_z\, m_3 - \tilde{u}_x\, m_7 + m_{15}. \tag{20}$$

5.3 Energy moment (degree 2, row 1)

$$k_1 = 19\, u^2\, m_0 + m_1 - 38\tilde{u}_x\, m_3 - 38\tilde{u}_y\, m_5 - 38\tilde{u}_z\, m_7. \tag{21}$$

5.4 Third-order moments (rows 16,17,18)

These polynomials have degree 3 and only involve monomials from the independent set (5), so no ambiguity:

$$\begin{aligned} k_{16} &= \tilde{u}_x(\tilde{u}_z^2 - \tilde{u}_y^2)\, m_0 + (\tilde{u}_y^2 - \tilde{u}_z^2)\, m_3 + 2\tilde{u}_x\tilde{u}_y\, m_5 - 2\tilde{u}_x\tilde{u}_z\, m_7 \\ &\quad - \tilde{u}_x\, m_{11} - 2\tilde{u}_y\, m_{13} + 2\tilde{u}_z\, m_{15} + m_{16}, \end{aligned} \tag{22}$$

$$\begin{aligned} k_{17} &= \tilde{u}_y(\tilde{u}_x^2 - \tilde{u}_z^2)\, m_0 - 2\tilde{u}_x\tilde{u}_y\, m_3 + (\tilde{u}_z^2 - \tilde{u}_x^2)\, m_5 + 2\tilde{u}_y\tilde{u}_z\, m_7 \\ &\quad + \tfrac{1}{2}\tilde{u}_y\, m_9 + \tfrac{1}{2}\tilde{u}_y\, m_{11} + 2\tilde{u}_x\, m_{13} - 2\tilde{u}_z\, m_{14} + m_{17}, \end{aligned} \tag{23}$$

$$\begin{aligned} k_{18} &= \tilde{u}_z(\tilde{u}_y^2 - \tilde{u}_x^2)\, m_0 + 2\tilde{u}_x\tilde{u}_z\, m_3 - 2\tilde{u}_y\tilde{u}_z\, m_5 + (\tilde{u}_x^2 - \tilde{u}_y^2)\, m_7 \\ &\quad - \tfrac{1}{2}\tilde{u}_z\, m_9 + \tfrac{1}{2}\tilde{u}_z\, m_{11} + 2\tilde{u}_y\, m_{14} - 2\tilde{u}_x\, m_{15} + m_{18}. \end{aligned} \tag{24}$$

5.5 Energy-flux moments (degree 3/4, rows 4,6,8) — version-dependent

Using the **lattice-reduced** polynomials $P_4^{\text{red}} = -4X + 5XY^2 + 5XZ^2$ (with $X^3 \to X$):

$$\begin{aligned} k_4 &= \left(-5\tilde{u}_x\tilde{u}_y^2 - 5\tilde{u}_x\tilde{u}_z^2 - \tfrac{24}{19}\tilde{u}_x\right) m_0 - \tfrac{10}{57}\tilde{u}_x\, m_1 \\ &\quad + 5(\tilde{u}_y^2 + \tilde{u}_z^2)\, m_3 + m_4 + 10\tilde{u}_x\tilde{u}_y\, m_5 + 10\tilde{u}_x\tilde{u}_z\, m_7 \\ &\quad + \tfrac{5}{3}\tilde{u}_x\, m_9 - 10\tilde{u}_y\, m_{13} - 10\tilde{u}_z\, m_{15}, \end{aligned} \tag{25}$$

and analogously for k_6 (cyclic: $x \to y$, $m_9 \to -\frac{5}{6}m_9 + \frac{5}{2}m_{11}$) and k_8 ($x \to z$, $m_9 \to -\frac{5}{6}m_9 - \frac{5}{2}m_{11}$).

5.6 Ghost/energy-squared moments (degree 4, rows 2,10,12) — version-dependent

Using the **lattice-reduced** polynomial P_2^{red}:

$$\begin{aligned} k_2 &= \left[21(\tilde{u}_x^2\tilde{u}_y^2 + \tilde{u}_x^2\tilde{u}_z^2 + \tilde{u}_y^2\tilde{u}_z^2) + \tfrac{116}{19}\, u^2\right] m_0 + \tfrac{14}{19}\, u^2\, m_1 + m_2 \\ &\quad + \left(-42\tilde{u}_x\tilde{u}_y^2 - 42\tilde{u}_x\tilde{u}_z^2 - \tfrac{8}{5}\tilde{u}_x\right) m_3 - \tfrac{42}{5}\tilde{u}_x\, m_4 \\ &\quad + \left(-42\tilde{u}_x^2\tilde{u}_y - 42\tilde{u}_y\tilde{u}_z^2 - \tfrac{8}{5}\tilde{u}_y\right) m_5 - \tfrac{42}{5}\tilde{u}_y\, m_6 \\ &\quad + \left(-42\tilde{u}_x^2\tilde{u}_z - 42\tilde{u}_y^2\tilde{u}_z - \tfrac{8}{5}\tilde{u}_z\right) m_7 - \tfrac{42}{5}\tilde{u}_z\, m_8 \\ &\quad + (-7\tilde{u}_x^2 + \tfrac{7}{2}\tilde{u}_y^2 + \tfrac{7}{2}\tilde{u}_z^2)\, m_9 + (-\tfrac{21}{2}\tilde{u}_y^2 + \tfrac{21}{2}\tilde{u}_z^2)\, m_{11} \\ &\quad + 84\tilde{u}_x\tilde{u}_y\, m_{13} + 84\tilde{u}_y\tilde{u}_z\, m_{14} + 84\tilde{u}_x\tilde{u}_z\, m_{15}. \end{aligned} \tag{26}$$

Rows 10 and 12 follow the same structure with coefficients derived from P_{10}^{red} and P_{12}^{red} (see Appendix A).

6 Inverse Shift Operator $\boldsymbol{T}^{-1}(\tilde{\boldsymbol{u}})$

6.1 Lattice-reduced version: exact inverse via sign flip

Theorem 6.1 (Group property)**.** *For the lattice-reduced polynomial basis, the shift operator satisfies:*

$$\boldsymbol{T}^{red}(\boldsymbol{u})\,\boldsymbol{T}^{red}(\boldsymbol{v}) = \boldsymbol{T}^{red}(\boldsymbol{u}+\boldsymbol{v}), \qquad \forall\, \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^3. \tag{27}$$

In particular, $\boldsymbol{T}^{red}(-\boldsymbol{u}) = [\boldsymbol{T}^{red}(\boldsymbol{u})]^{-1}$.

Proof. The lattice-reduced polynomials use only the 19 independent monomials (5). Denote $\boldsymbol{X} = (X, Y, Z)$. The key observation is that the map $\boldsymbol{X} \mapsto \boldsymbol{X} - \boldsymbol{u}$ acts as a polynomial automorphism on the ring $\mathbb{R}[X, Y, Z]/\mathcal{I}$, where $\mathcal{I}$ is the ideal generated by $\{X^4 - X^2, Y^4 - Y^2, Z^4 - Z^2, X^3 - X, Y^3 - Y, Z^3 - Z\}$.

In this quotient ring, the shift $(X, Y, Z) \to (X - u_x, Y - u_y, Z - u_z)$ followed by $(X, Y, Z) \to (X - v_x, Y - v_y, Z - v_z)$ is the same as a single shift $(X, Y, Z) \to (X - u_x - v_x, Y - u_y - v_y, Z - u_z - v_z)$. Since $\boldsymbol{M}^{-1}(0)$ maps into the quotient ring, the composition $\boldsymbol{T}(\boldsymbol{u})\,\boldsymbol{T}(\boldsymbol{v})$ equals $\boldsymbol{T}(\boldsymbol{u} + \boldsymbol{v})$.

This has been verified symbolically and numerically to machine precision. □

Corollary 6.1. *The inverse shift operator $\boldsymbol{T}^{-1}(\boldsymbol{u})$ is obtained by replacing $\boldsymbol{u} \to -\boldsymbol{u}$ in all expressions. Concretely, for the lattice-reduced version, the inverse $\boldsymbol{m} = \boldsymbol{T}^{-1}(\boldsymbol{u})\,\boldsymbol{k}$ is obtained by:*

1. *Flipping the sign of all odd-power terms in $\tilde{u}_x, \tilde{u}_y, \tilde{u}_z$.*
2. *Exchanging the roles of $\boldsymbol{m}$ and $\boldsymbol{k}$.*

For example, from (13)*: $k_3 = -\tilde{u}_x\, m_0 + m_3$ becomes $m_3 = +\tilde{u}_x\, k_0 + k_3$.*

6.2 Dubois continuous version: approximate inverse

Proposition 6.1. *For the Dubois continuous polynomial basis, the group property fails:*

$$\boldsymbol{T}^{cont}(\boldsymbol{u})\,\boldsymbol{T}^{cont}(-\boldsymbol{u}) \neq \boldsymbol{I}. \tag{28}$$

The error is $O(|\boldsymbol{u}|^2)$ and is concentrated in rows $\{2, 4, 6, 8, 10, 12\}$.

Proof. The continuous polynomial $r^4 = (X^2 + Y^2 + Z^2)^2$ has degree 4 in the unreduced ring. At shifted points $\boldsymbol{e}_i - \boldsymbol{u}$, expanding $(e_{ix} - u_x)^4$ introduces the monomial u_x^4, which is *not* in the lattice-reduced ring (it would be u_x^2 under lattice reduction, but u_x is continuous, not a lattice variable).

More precisely, $\boldsymbol{M}^{-1}(0)$ was constructed from the 19 lattice-independent monomials. The continuous polynomial $P_k^{\text{cont}}(\boldsymbol{e}_j - \boldsymbol{u})$ introduces "ghost" monomials $e_{jx}^4 - 4e_{jx}^3 u_x + \cdots$ that project onto the 19-dimensional lattice space through aliasing: $e_{jx}^4 \to e_{jx}^2$. This aliasing breaks the automorphism property.

Numerical verification: at $\boldsymbol{u} = (0.12, -0.07, 0.05)$,

$$\|\boldsymbol{T}^{\text{cont}}(\boldsymbol{u})\,\boldsymbol{T}^{\text{cont}}(-\boldsymbol{u}) - \boldsymbol{I}\|_F = 0.345.$$

□

7 Comparison of the Two Versions

7.1 What is identical

Corollary 7.1 (Dubois et al. 2015, Corollary 2.1)**.** *The second-order equivalent partial differential equations (and hence the Navier–Stokes viscosity) are independent of the polynomial extension choice. Specifically, the stress-mode central moments (rows 9, 11, 13, 14, 15) are* identical *between both versions:*

$$[\boldsymbol{T}^{cont}(\boldsymbol{u})]_{k,:} = [\boldsymbol{T}^{red}(\boldsymbol{u})]_{k,:}, \qquad k \in \{0, 1, 3, 5, 7, 9, 11, 13, 14, 15, 16, 17, 18\}. \tag{29}$$

This follows because these rows correspond to polynomials of degree ≤ 3 that involve only the 19 independent monomials — no degree-4 ambiguity exists.

7.2 What differs

The 6 rows $\{2, 4, 6, 8, 10, 12\}$ differ. These are the "starred" ghost modes (energy-squared, energy-flux, energy×stress). The difference in $\boldsymbol{T}$ is an additive correction $\Delta\boldsymbol{T}$:

$$\boldsymbol{T}^{\text{cont}} = \boldsymbol{T}^{\text{red}} + \Delta\boldsymbol{T}, \qquad \text{where } \Delta\boldsymbol{T} \text{ has nonzero entries only in rows } \{2, 4, 6, 8, 10, 12\}. \tag{30}$$

7.3 Correction structure

The correction $\Delta \boldsymbol{T}$ can be expressed in terms of the "correction polynomials":

$$\delta(\alpha) = \alpha^2(\alpha^2 - 1), \qquad \gamma(\alpha) = \alpha(\alpha^2 - 1). \tag{31}$$

These vanish on the lattice ($\alpha \in \{-1, 0, 1\} \Rightarrow \delta = \gamma = 0$) but are nonzero at shifted points.

The correction terms are proportional to $\delta(\tilde{u}_x), \gamma(\tilde{u}_x)$, etc., and thus scale as $O(\tilde{u}^2)$ for small $\tilde{u}$. In particular, at $\tilde{u} = 0$, $\Delta \boldsymbol{T} = 0$ (both versions degenerate to the identity).

7.4 Summary table

Property	Lattice-Reduced	Dubois Continuous
Group: $\boldsymbol{T}(\boldsymbol{u})\boldsymbol{T}(-\boldsymbol{u}) = \boldsymbol{I}$	**Exact**	**Fails** ($O(u^2)$)
Morphism: $\boldsymbol{T}(\boldsymbol{u}+\boldsymbol{v}) = \boldsymbol{T}(\boldsymbol{u})\boldsymbol{T}(\boldsymbol{v})$	**Exact**	**Fails** ($O(u^2)$)
Inverse: $\boldsymbol{T}^{-1} = \boldsymbol{T}(-\boldsymbol{u})$	**Yes**	**No** (requires matrix inv.)
Stress CMs	Identical	Identical
N-S viscosity	Identical	Identical
Ghost/energy CMs	Version A	Version B $\neq$ A
Stability	✓	Possibly different

8 Physical Interpretation

The MRT-CM collision (1) relaxes the non-equilibrium central moments with rates s_k. The conserved central moments (k_0, k_3, k_5, k_7) have $s_k = 0$ (not relaxed). The stress central moments $(k_9, k_{11}, k_{13}, k_{14}, k_{15})$ are relaxed with rate s_ν that determines the kinematic viscosity:

$$\nu = \frac{1}{3}\left(\frac{1}{s_\nu} - \frac{1}{2}\right) \Delta x^2/\Delta t. \tag{32}$$

Since the stress CMs are identical between both versions (Corollary 7.1), the physical viscosity is unchanged regardless of which polynomial extension is used.

The ghost modes (rows 2, 4, 6, 8, 10, 12) affect only higher-order error terms and stability margins. The lattice-reduced version, with its exact algebraic structure, is expected to provide more consistent stability behavior, particularly at higher Mach numbers where $|\Delta \boldsymbol{T}|$ becomes significant.

9 Implementation Notes

9.1 Efficient CUDA implementation

The shift operator does **not** require a 19×19 matrix–vector multiply. Both $\boldsymbol{T}(\boldsymbol{u})\,\boldsymbol{m}$ and $\boldsymbol{T}^{-1}(\boldsymbol{u})\,\boldsymbol{k}$ can be evaluated as hand-coded arithmetic expressions (Section 5), requiring approximately:

	Multiplications	Additions
$\boldsymbol{T}\,\boldsymbol{m}$ (forward)	~ 120	~ 100
$\boldsymbol{T}^{-1}\,\boldsymbol{k}$ (inverse)	~ 120	~ 100

This is far cheaper than a full 19×19 matrix multiply ($361 \times 2 = 722$ FLOPs).

9.2 Compile-time switch

In our GILBM code, a compile-time macro selects the version:

```
#define USE_DUBOIS_CONTINUOUS  0  // 0 = lattice-reduced, 1 = Dubois continuous
```

The Dubois version is implemented as an additive correction to the lattice-reduced version:

```
raw_to_central_dH_dubois(m, ux, uy, uz, k):
    raw_to_central_dH(m, ux, uy, uz, k);     // lattice-reduced base
    k[2]  += correction_row2(m, ux, uy, uz);  // additive DeltaT
    k[4]  += correction_row4(m, ux, uy, uz);
    ...
    k[12] += correction_row12(m, ux, uy, uz);
```

This structure minimizes code duplication and allows direct comparison.

10 Conclusion

We have presented a complete derivation of the central moment shift operator $\boldsymbol{T}(\tilde{\boldsymbol{u}})$ for D3Q19 MRT lattice Boltzmann methods, filling a gap in the literature where Dubois et al. (2015) only provided explicit D2Q9 expressions.

The key insight is that the d'Humières D3Q19 polynomial basis has a **non-unique continuous extension** due to the lattice identity $x^4 = x^2$ (for $x \in \{-1, 0, 1\}$). We identified two natural choices:

1. The "Dubois continuous" form, which preserves $r^4 = (r^2)^2$ as a true degree-4 polynomial.
2. The "lattice-reduced" form, which uses only 19 independent monomials by applying $x^4 \to x^2$, $x^3 \to x$.

The lattice-reduced form is algebraically superior: it satisfies the group property $\boldsymbol{T}(\boldsymbol{u})\,\boldsymbol{T}(-\boldsymbol{u}) = \boldsymbol{I}$ exactly, and provides an explicit inverse $\boldsymbol{T}^{-1}(\boldsymbol{u}) = \boldsymbol{T}(-\boldsymbol{u})$ without matrix inversion. Both forms yield identical stress-mode central moments and hence the same Navier–Stokes viscosity.

A Full Expressions for All Remaining Rows

A.1 Row 10: $3\pi_{xx}$ (energy $\times$ stress)

$$
\begin{aligned}
k_{10} = {} & \left[3(\tilde{u}_x^2\tilde{u}_y^2 + \tilde{u}_x^2\tilde{u}_z^2) - 6\tilde{u}_y^2\tilde{u}_z^2 - \tfrac{16}{19}\tilde{u}_x^2 + \tfrac{8}{19}\tilde{u}_y^2 + \tfrac{8}{19}\tilde{u}_z^2\right] m_0 \\
& + \tfrac{1}{19}(2\tilde{u}_x^2 - \tilde{u}_y^2 - \tilde{u}_z^2)\, m_1 \\
& + (-6\tilde{u}_x\tilde{u}_y^2 - 6\tilde{u}_x\tilde{u}_z^2 + \tfrac{16}{5}\tilde{u}_x)\, m_3 - \tfrac{6}{5}\tilde{u}_x\, m_4 \\
& + (-6\tilde{u}_x^2\tilde{u}_y + 12\tilde{u}_y\tilde{u}_z^2 - \tfrac{8}{5}\tilde{u}_y)\, m_5 + \tfrac{3}{5}\tilde{u}_y\, m_6 \\
& + (-6\tilde{u}_x^2\tilde{u}_z + 12\tilde{u}_y^2\tilde{u}_z - \tfrac{8}{5}\tilde{u}_z)\, m_7 + \tfrac{3}{5}\tilde{u}_z\, m_8 \\
& + (-\tilde{u}_x^2 + 2\tilde{u}_y^2 + 2\tilde{u}_z^2)\, m_9 + m_{10} + (3\tilde{u}_y^2 - 3\tilde{u}_z^2)\, m_{11} \\
& + 12\tilde{u}_x\tilde{u}_y\, m_{13} - 24\tilde{u}_y\tilde{u}_z\, m_{14} + 12\tilde{u}_x\tilde{u}_z\, m_{15} \\
& + 9\tilde{u}_y\, m_{17} - 9\tilde{u}_z\, m_{18}.
\end{aligned} \tag{33}
$$

A.2 Row 12: π_{ww} (energy $\times$ stress)

$$
\begin{aligned}
k_{12} = {} & \left[3\tilde{u}_x^2(\tilde{u}_y^2 - \tilde{u}_z^2) + \tfrac{8}{19}(\tilde{u}_z^2 - \tilde{u}_y^2)\right] m_0 + \tfrac{1}{19}(\tilde{u}_y^2 - \tilde{u}_z^2)\, m_1 \\
& - 6\tilde{u}_x(\tilde{u}_y^2 - \tilde{u}_z^2)\, m_3 \\
& + \tfrac{2}{5}\tilde{u}_y(4 - 15\tilde{u}_x^2)\, m_5 - \tfrac{3}{5}\tilde{u}_y\, m_6 \\
& + \tfrac{2}{5}\tilde{u}_z(15\tilde{u}_x^2 - 4)\, m_7 + \tfrac{3}{5}\tilde{u}_z\, m_8 \\
& + (\tilde{u}_y^2 - \tilde{u}_z^2)\, m_9 + 3\tilde{u}_x^2\, m_{11} + m_{12} \\
& + 12\tilde{u}_x\tilde{u}_y\, m_{13} - 12\tilde{u}_x\tilde{u}_z\, m_{15} \\
& - 6\tilde{u}_x\, m_{16} + 3\tilde{u}_y\, m_{17} + 3\tilde{u}_z\, m_{18}.
\end{aligned} \tag{34}
$$

A.3 Row 6: q_y (energy flux, y-direction)

$$
\begin{aligned}
k_6 = {} & (-5\tilde{u}_x^2\tilde{u}_y - 5\tilde{u}_y\tilde{u}_z^2 - \tfrac{24}{19}\tilde{u}_y)\, m_0 - \tfrac{10}{57}\tilde{u}_y\, m_1 \\
& + 10\tilde{u}_x\tilde{u}_y\, m_3 + 5(\tilde{u}_x^2 + \tilde{u}_z^2)\, m_5 + m_6 + 10\tilde{u}_y\tilde{u}_z\, m_7 \\
& - \tfrac{5}{6}\tilde{u}_y\, m_9 + \tfrac{5}{2}\tilde{u}_y\, m_{11} - 10\tilde{u}_x\, m_{13} - 10\tilde{u}_z\, m_{14}.
\end{aligned} \tag{35}
$$

A.4 Row 8: q_z (energy flux, z-direction)

$$
\begin{aligned}
k_8 = {} & (-5\tilde{u}_x^2\tilde{u}_z - 5\tilde{u}_y^2\tilde{u}_z - \tfrac{24}{19}\tilde{u}_z)\, m_0 - \tfrac{10}{57}\tilde{u}_z\, m_1 \\
& + 10\tilde{u}_x\tilde{u}_z\, m_3 + 10\tilde{u}_y\tilde{u}_z\, m_5 + 5(\tilde{u}_x^2 + \tilde{u}_y^2)\, m_7 + m_8 \\
& - \tfrac{5}{6}\tilde{u}_z\, m_9 - \tfrac{5}{2}\tilde{u}_z\, m_{11} - 10\tilde{u}_y\, m_{14} - 10\tilde{u}_x\, m_{15}.
\end{aligned} \tag{36}
$$

B Dubois Correction Terms $\Delta \boldsymbol{T}$

The additive correction $\Delta \boldsymbol{T}$ satisfies $\boldsymbol{T}^{\text{cont}}\, \boldsymbol{m} = \boldsymbol{T}^{\text{red}}\, \boldsymbol{m} + \Delta \boldsymbol{T}\, \boldsymbol{m}$. Only rows $\{2, 4, 6, 8, 10, 12\}$ are nonzero, and each row's correction depends only on $m_0, m_1, m_3, m_5, m_7, m_9, m_{11}$ (at most 7 moment entries):

Row 2:

$$
\begin{aligned}
\Delta k_2 = {} & \left[\tfrac{21}{2}\tilde{u}_x^4 + \tfrac{861}{38}\tilde{u}_x^2 + \tfrac{21}{2}\tilde{u}_y^4 + \tfrac{861}{38}\tilde{u}_y^2 + \tfrac{21}{2}\tilde{u}_z^4 + \tfrac{861}{38}\tilde{u}_z^2\right] m_0 \\
& + \tfrac{21}{19}(\tilde{u}_x^2 + \tilde{u}_y^2 + \tilde{u}_z^2)\, m_1 \\
& + (-42\tilde{u}_x^3 - 21\tilde{u}_x)\, m_3 + (-42\tilde{u}_y^3 - 21\tilde{u}_y)\, m_5 + (-42\tilde{u}_z^3 - 21\tilde{u}_z)\, m_7 \\
& + (21\tilde{u}_x^2 - \tfrac{21}{2}\tilde{u}_y^2 - \tfrac{21}{2}\tilde{u}_z^2)\, m_9 + (\tfrac{63}{2}\tilde{u}_y^2 - \tfrac{63}{2}\tilde{u}_z^2)\, m_{11}.
\end{aligned} \tag{37}
$$

Row 4:

$$
\Delta k_4 = (-5\tilde{u}_x^3 - \tfrac{55}{19}\tilde{u}_x)\, m_0 - \tfrac{5}{19}\tilde{u}_x\, m_1 + 15\tilde{u}_x^2\, m_3 - 5\tilde{u}_x\, m_9. \tag{38}
$$

Row 6:

$$
\Delta k_6 = (-5\tilde{u}_y^3 - \tfrac{55}{19}\tilde{u}_y)\, m_0 - \tfrac{5}{19}\tilde{u}_y\, m_1 + 15\tilde{u}_y^2\, m_5 + \tfrac{5}{2}\tilde{u}_y\, m_9 - \tfrac{15}{2}\tilde{u}_y\, m_{11}. \tag{39}
$$

Row 8:

$$
\Delta k_8 = (-5\tilde{u}_z^3 - \tfrac{55}{19}\tilde{u}_z)\, m_0 - \tfrac{5}{19}\tilde{u}_z\, m_1 + 15\tilde{u}_z^2\, m_7 + \tfrac{5}{2}\tilde{u}_z\, m_9 + \tfrac{15}{2}\tilde{u}_z\, m_{11}. \tag{40}
$$

Row 10:

$$
\begin{aligned}
\Delta k_{10} = {} & (6\tilde{u}_x^4 + \tfrac{246}{19}\tilde{u}_x^2 - 3\tilde{u}_y^4 - \tfrac{123}{19}\tilde{u}_y^2 - 3\tilde{u}_z^4 - \tfrac{123}{19}\tilde{u}_z^2)\, m_0 \\
& + \tfrac{1}{19}(12\tilde{u}_x^2 - 6\tilde{u}_y^2 - 6\tilde{u}_z^2)\, m_1 \\
& + (-24\tilde{u}_x^3 - 12\tilde{u}_x)\, m_3 + (12\tilde{u}_y^3 + 6\tilde{u}_y)\, m_5 + (12\tilde{u}_z^3 + 6\tilde{u}_z)\, m_7 \\
& + (12\tilde{u}_x^2 + 3\tilde{u}_y^2 + 3\tilde{u}_z^2)\, m_9 + (-9\tilde{u}_y^2 + 9\tilde{u}_z^2)\, m_{11}.
\end{aligned} \tag{41}
$$

Row 12:

$$\begin{aligned}\Delta k_{12} = &(3\tilde{u}_y^4 + \tfrac{123}{19}\tilde{u}_y^2 - 3\tilde{u}_z^4 - \tfrac{123}{19}\tilde{u}_z^2)\, m_0 + \tfrac{1}{19}(6\tilde{u}_y^2 - 6\tilde{u}_z^2)\, m_1 \\ &+ (-12\tilde{u}_y^3 - 6\tilde{u}_y)\, m_5 + (12\tilde{u}_z^3 + 6\tilde{u}_z)\, m_7 \\ &+ (-3\tilde{u}_y^2 + 3\tilde{u}_z^2)\, m_9 + (9\tilde{u}_y^2 + 9\tilde{u}_z^2)\, m_{11}. \end{aligned} \tag{42}$$

References

[1] F. Dubois, T. Fevrier, and B. Graille, "Lattice Boltzmann schemes with relative velocities," *Commun. Comput. Phys.*, vol. 17, no. 4, pp. 1088–1112, 2015. arXiv:1312.3297.

[2] D. d'Humières, I. Ginzburg, M. Krafczyk, P. Lallemand, and L.-S. Luo, "Multiple-relaxation-time lattice Boltzmann models in three dimensions," *Phil. Trans. R. Soc. Lond. A*, vol. 360, pp. 437–451, 2002.

[3] M. Geier, A. Greiner, and J. G. Korvink, "Cascaded digital lattice Boltzmann automata for high Reynolds number flow," *Phys. Rev. E*, vol. 73, p. 066705, 2006.

[4] T. Imamura, K. Suzuki, T. Nakamura, and M. Yoshida, "Acceleration of steady-state lattice Boltzmann simulations on non-uniform mesh using local time step method," *J. Comput. Phys.*, vol. 202, pp. 645–663, 2005.